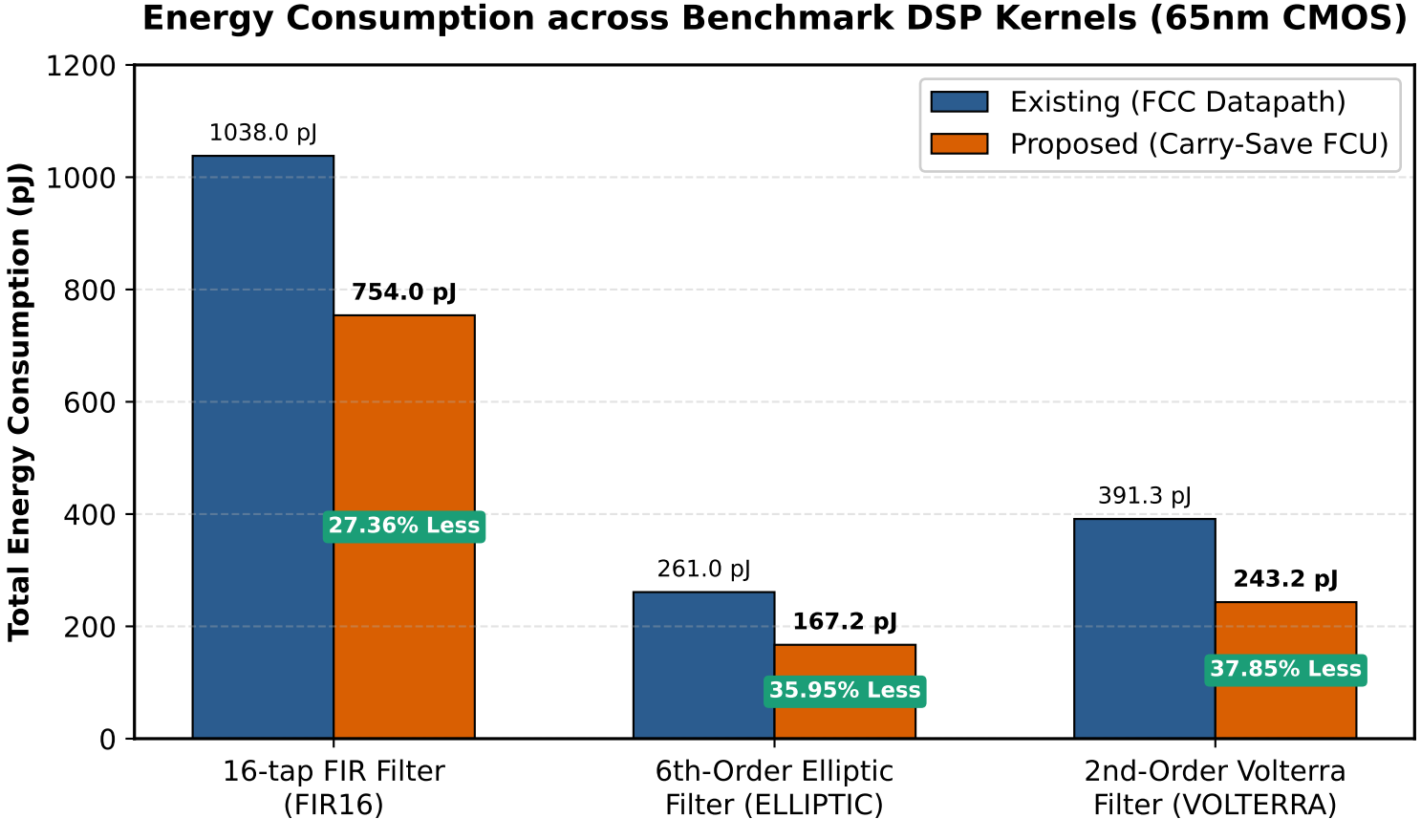
Energy Consumption across Benchmark DSP Kernels (65nm CMOS)
Total Energy Consumption (pJ)
0
200
400
600
800
1000
1200
Existing (FCC Datapath)
Proposed (Carry-Save FCU)
1038.0 pJ
754.0 pJ
27.36% Less
261.0 pJ
167.2 pJ
35.95% Less
391.3 pJ
243.2 pJ
37.85% Less
16-tap FIR Filter
(FIR16)
6th-Order Elliptic
Filter (ELLIPTIC)
2nd-Order Volterra
Filter (VOLTERRA)